

Summary of Findings: Visualization Task

Alt Mobility – Data Analyst Intern Assignment

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Summary of Findings

Part 1: Key Insights from SQL Queries

1. Order and Sales Analysis

Majority of orders fall under delivered and shipped categories, indicating effective fulfillment processes.

Delivered orders generate the highest revenue, as expected, while pending orders contribute less and may indicate customer drop-offs or internal delays.

The average order value provides a baseline for revenue forecasting and helps identify potential upselling opportunities.

2. Customer Behaviour Analysis

A significant portion of customers are repeat buyers, showing a strong retention trend.

Several customers have a long history of transactions, indicating loyalty and product satisfaction.

Customers placing orders within shorter intervals could be high-value targets for subscription-based or loyalty programs.

3. Payment Status Analysis

Over 90% of payments are completed, but there are some failed payments.

Failed payments represent lost revenue and should be minimized through improved payment systems or customer support intervention.

4. Order Details Integration

Joining orders and payments tables revealed a small number of orders with missing or failed payment records.

This integration helps identify fulfillment vs. financial mismatches for further reconciliation or fraud detection.

Part 2: Customer Retention Analysis

What Was Done:

Conducted cohort analysis using customer first order month as the base.

Calculated retention rates as the percentage of customers who returned in subsequent months.

Visualized data using a retention heatmap.

Key Insights:

Retention is strongest in the first 1–2 months after the initial order, then gradually declines.

Some cohorts show surprisingly high retention in later months, possibly due to seasonality, campaigns, or onboarding quality.

Retention decay indicates opportunities for better engagement in months 3+ (e.g., reminder emails, discounts).

Visualization Description:

Rows = Cohort months (e.g., Jan 2023)

Columns = Months since first order (cohort index: 0 = same month, 1 = next, etc.)

Each cell = % of retained customers from that cohort

Visualized as a heatmap for quick trend detection.

Recommendations for Alt Mobility

- Target high-LTV customers identified through order frequency and revenue.
- Launch re-engagement campaigns after Month 2 to counter retention drop-off.
- Improve payment reliability to reduce transaction failure rates.
- Use cohort trends to evaluate onboarding changes and campaign effectiveness.
- Expand use of data dashboards to monitor KPIs like retention, revenue, and active users in real-time.